

JTAG Enumeration for Security Research on the Zynq UltraScale+ MPSoC

Volume 3 — What Enumeration Reveals (and How to Use It)

Enumeration over JTAG is a **security-configuration assessment**: read the registers where ZynqMP's security controls live, and their values tell you which controls are on. This volume is the field guide. Each indicator category lists the exact attributes and **where to find them in a register dump** (block · address · bits), how the dev-board baseline compares to a hardened part, and what the observed state enables.

The reference target is the ZCU102 (XCZU9EG, S/N 210308BD8D4D): factory silicon with **no security provisioned** — the **all-open baseline**, where every control reads OFF/unprovisioned. The payoff is comparison: point the same enumeration at another (hardened) board and the deviations from this baseline are the story. The exhaustive flat catalog of every attribute is [../11-security-attributes.md](#); this volume is the interpretive companion to it.

1. How to read this volume

Every category has the same two parts:

- **Attributes & locations** — a table of *what* each attribute is, *where* it lives (BLOCK.REGISTER · 0xADDRESS · bits), its value on this dev board, and what a hardened part shows instead. The address+bits column is what lets you locate the attribute in a raw register dump.
- **What it enables** — the concrete techniques the observed state allows, with prerequisites and limits.

Read every value against the **hardening continuum** — a reading that's unremarkable on a dev kit is significant on a production device, and vice-versa:

Factory	→	Dev kit	→	Partial hardening	→	Production hardened
unblown		blown but		blown + partial		all gates closed,
fuses		mutable		locking		keys baked, JTAG off

Scope. This assumes the debug threat model — JTAG/DAP is reachable. On hardened silicon that access is itself closed (JTAG_DIS or debug authentication), so “JTAG can read X” only applies where JTAG is open by design or by misconfiguration. No fault injection, side-channel, or supply-chain pre-positioning.

The fastest read on a new board is the **Security Posture Summary** (§10) — a one-row-per-control checklist that flips OFF/dev → ON/provisioned.

2. Silicon identity

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
Device part / family	CSU.IDCODE · 0xFFCA0040	0x24738093 (XCZU9EG)	identifies die + revision
Silicon revision	CSU.VERSION · 0xFFCA0044 (PS_VERSION)	3 (production)	0/1 = ES1/ES2 sample
Per-chip unique ID	EFUSE.DNA_0..2 · 0xFFCC100C–1014	0x40000000_0170CFA7_14804345	read-gated

What it enables

- *Pre-engagement recon.* Exact die + revision tells you which errata apply, which boot-image format the BootROM expects, and what prior exploit work exists.
- *Engineering-sample detection.* An ES revision in a shipped product is worth escalating — ES silicon often carries security errata fixed in production steppings.
- *Fleet fingerprinting.* DNA is unique per chip and readable from the eFuse shadow; across a fleet it builds a DNA→identity table. DNA readable on a part that should lock it is itself a finding.
- *Variant inference.* If the variant table claims a subsystem (e.g. VCU) but its base address times out — or vice-versa — the package bond differs from the catalog.

3. Secure-debug exposure (the most consequential indicator)

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
APU secure invasive debug	CSU.JTAG_DAP_CFG · 0xFFCA003C · bit 2 (SPIDEN)	1 (open)	0
APU secure trace	0xFFCA003C · bit 3 (SPNIDEN)	1 (open)	0
APU non-secure debug	0xFFCA003C · bit 0 (DBGEN)	1	0 or 1
APU non-secure trace	0xFFCA003C · bit 1 (NIDEN)	1	0
RPU debug / trace	0xFFCA003C · bits 4,5 (RPU DBGEN/NIDEN)	1	0
DAP JTAG path	CSU.JTAG_SEC · 0xFFCA0038 · bits [2:0] (DAP_SEC)	0b111 (open)	not 0b111
PL-TAP JTAG path	0xFFCA0038 · bits [5:3] (PLTAP_SEC)	0b111 (open)	not 0b111

Attribute	Where (block · address · bits)	This board	Hardened part
PMU JTAG path	0xFFCA0038 · bits [8:6] (PMU_SEC)	0b000 (gated)	gated

The whole JTAG_DAP_CFG reads 0xFF and JTAG_SEC reads 0x3F on this board — maximum debug exposure. Each JTAG_SEC field needs the magic 0b111 to unlock (anti-glitch: a single-bit flip can't open it).

What it enables

- *TrustZone inspection (SPIDEN=1)*. Halt the A53 in EL3 and read everything the secure world holds — keys in OCM/DDR, the EL3 monitor, attestation material.
- *Secure-world key extraction*. Keys the BootROM/ATF stage into secure RAM are directly readable; eFuse-shadow keys too, if not separately gated.
- *EL3 implant*. Write a secure region, redirect the secure vector table into it, resume — the EL3 monitor then runs attacker code on every SMC.
- *Secure trace (SPNIDEN=1, SPIDEN=0)*. Watch secure execution live without halt control — map the SMC surface, see when/where keys load.
- *Non-secure debug (DBGEN=1, SPIDEN=0)*. Halt and patch the kernel/RTOS, disable checks, escalate privilege.
- *Pre-attack confirmation*. SPIDEN=1 makes expensive attack development (fault injection, supply-chain) unnecessary — confirm the easy path first.

4. eFuse secure-boot policy

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
RSA boot authentication	EFUSE.SEC_CTRL · bits [25:11] (RSA_EN)	0 (off)	15-bit magic = enforced
Encrypt-only boot	0xFFCC1058 · bit 2 (ENC_ONLY)	0	1
JTAG disable	0xFFCC1058 · bit 5 (JTAG_DIS)	0	1
Secure lockdown	0xFFCC1058 · bit 10 (SEC_LOCK)	0	1
PPK0 public-key hash	EFUSE.PPK0_0..11 · 0xFFCC10A0–10CC	all-zero	non-zero (root of trust)
PPK1 public-key hash	EFUSE.PPK1_0..11 · 0xFFCC10D0–10FC	all-zero	non-zero

Attribute	Where (block · address · bits)	This board	Hardened part
PPK0 / PPK1 revoked	0xFFCC1058 · [28:27] / [31:30] (PPK_INVLD)	0	2-bit magic = revoked

Field widths are anti-glitch defenses (RSA_EN is 15 bits, each PPK_INVLD 2 bits) — a single-bit fault can't enable a bypass or revoke a key, so a *partial* magic pattern is a red flag.

What it enables

- *Boot-chain enforcement profile.* SEC_CTRL = 0 ⇒ unsigned/unencrypted boot accepted (full boot-chain control). RSA magic only ⇒ signature required. RSA + ENC_ONLY ⇒ signed-encrypted only.
- *Image substitution.* On a low/partial-enforcement target, swap in attacker firmware. A partial profile (RSA_EN set, SEC_LOCK clear) is still mutable.
- *AES key-extraction feasibility.* AES_RDLC = 0 means the eFuse AES key is readable — recover it and all “encrypted” images decrypt offline (see §5).
- *PPK-revocation bypass.* A partial PPK_INVLD pattern can mean an incomplete key rotation — the BootROM may still trust a key meant to be revoked.
- *Fleet outlier hunting.* Inventory SEC_CTRL across a product line; the less-hardened units are the higher-value targets.

5. Key material and confidentiality

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
AES key-slot population	CSU.AES_STATUS · 0xFFCA1000 · bits 8–11 (per-slot * _ZERO)	0xF00 (all slots empty)	a slot's bit clear = key loaded
eFuse AES key present	EFUSE.EFUSE_AES_CRC · 0xFFCC0048	0 (absent)	non-zero (key burned)
AES key read-lock	EFUSE.SEC_CTRL · 0xFFCC1058 · bit 0 (AES_RDLC)	0 (readable)	1
AES key write-lock	0xFFCC1058 · bit 1 (AES_WRLK)	0	1
PUF provisioned	EFUSE.PUF_CHASH · 0xFFCC1050	0 (unprovisioned)	non-zero

What it enables

- *Key-model determination.* The combination says whether the device decrypts boot with an eFuse AES key, a BBRAM key, or a PUF-derived key — which decides the extraction approach.
- *Direct key disclosure.* “Key present (AES_CRC $\neq$ 0) + AES_RDLK = 0” is the flag: the boot-decryption key is unprotected from a debug read. Recover it → decrypt every encrypted image for that device, or forge new ones.
- *PUF parts.* A PUF-provisioned part derives the key at boot rather than storing it; extraction shifts to helper-data handling, but the enumeration still tells you which class of attack applies.

6. Anti-tamper

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
Tamper response policy	CSU.CSU_TAMPER_0..12 · 0xFFCA5004–5034	All-zero (disarmed)	non-zero per armed source
Tamper events latched	CSU.TAMPER_STATUS · 0xFFCA5000	0 (none)	non-zero = a source fired
Software tamper trigger	CSU.CSU_TAMPER_TRIG0 · 0xFFCA0014		—

Each CSU_TAMPER_n register wires one tamper source to a response (system reset / secure lockdown / BBRAM + key zeroize).

What it enables

- *Tamper-aware operation.* The config registers show which physical / voltage / temperature / JTAG-activity events are armed — i.e. what *not* to trip on a live target, since a response may zeroize keys or lock the device down.
- *Response-gap hunting.* A device that arms some sources but leaves the consequential ones (key zeroize) unconfigured is a softer target.
- *Forensics.* Latched TAMPER_STATUS bits on a fielded unit are evidence a source already fired.

7. Memory and peripheral TrustZone

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
DDR TrustZone (ch. 0–5)	DDR_XMPU0..5.CTRL/LOCK · 0x00000000...0x00000000	0x00000000 · permit	default-region + LOCK set
FPD TrustZone	FPD_XMPU.CTRL/LOCK · 0x00000000	—	configured
OCM TrustZone	OCM_XMPU.CTRL · 0x00000000	0x3	regions configured
Peripheral protection	XPPU.CTRL/ISR · 0x00000000	enabled, no violations	apertures locked down

ZynqMP enforces memory/peripheral TrustZone with XMPU (ranges) and XPPU (apertures) — there is no single TZASC.

What it enables

- *Cross-master reachability.* XMPU/XPPU config tells you whether a foothold on one master (PCIe, USB, a released R5) can reach secure DDR/OCM. Default/open XMPU means a compromised master can read secure memory.
- *Misconfiguration as a path.* A permissive region — or one with LOCK clear, so reconfigurable — is a direct route into protected memory.
- *PL-TAP exposure.* With PLTAP_SEC = 0b111 (§3) the FPGA JTAG TAP is reachable — read/modify configuration memory, extract or load bitstreams.

8. Boot and runtime state

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
Boot mode	BOOT_MODE_USER · 0xFF5E0200 · bits [3:0]	0 (JTAG idle)	SD/QSPI/eMMC/USB
Multiboot search offset	CSU.CSU_MULTI_BOOT · 0xFFCA0010	0	image-chaining offset
Auth / encrypt status	CSU.CSU_STATUS · 0xFFCA0000 (BOOT_AUTH/BOOT_ENC)	0	set after authenticated boot
Boot artifacts (when booted)	OCM 0xFFFFC0000 (FSBL), DDR 0x08000000 (U-Boot), 0x00200000 (kernel), ...	n/a in JTAG-idle	present + dumpable

What it enables

- *Strategy by state.* JTAG-idle is a clean slate (only BootROM ran). A booted device has FSBL/U-Boot/kernel up and secrets in DDR — larger but state-dependent.
 - *Live firmware extraction.* Once an artifact is located, JTAG reads it straight from physical memory, bypassing software read-protection (`dump_image uboot.bin 0x08000000 0x100000`). Version strings in the dump drive version-specific targeting.
 - *ATF/BL31 patch.* If ATF is at `0xFFFFEA000` (OCM) or in DDR, the EL3 monitor is readable and — with SPIDEN + write access — patchable.
 - *MULTI_BOOT image switching.* Writing `CSU_MULTI_BOOT` + reset changes which image the BootROM loads, given boot-media write access.
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9. Power, clock, reset

Attributes & locations

Attribute	Where (block · address · bits)	This board	Hardened part
Powered domains	<code>PMU_GLOBAL.PWR_STATE</code> · <code>0xFFD80100</code>	Only BootROM-era domains up	per running stack
Peripheral clocks	per-peripheral <code>*_REF_CTRL</code> (e.g. <code>0xFF5E0050+</code>)	most gated in idle	per running stack
Reset state	<code>RST_FPD_APU</code> · <code>0xFD1A0104</code> , <code>RST_LPD_I0U2</code> · <code>0xFF5E0238</code> , ...	cores/peripherals in reset	cleared per running stack
A53 reset-vector	<code>APU.RVBARADDR0L</code> · <code>0xFD5C0040</code>	—	(used by the release recipe)

What it enables

- *The A53 release primitive.* Release A53 core 0 over raw JTAG with no FSBL: set the landing pad in `RVBARADDR0L`, clear the `RST_FPD_APU` reset bits (ACPU0 / L2 / PWRON), halt — the core lands in EL3. From there: load code, read any memory, modify the EL3 monitor.
 - *Peripheral bring-up.* Clear a peripheral's reset bit + configure its `*_REF_CTRL` to bring it online (e.g. UART for console capture, or to drive a device).
 - *Live attack-surface map.* A peripheral both clocked and out-of-reset (UART, GEM, USB) is directly targetable — GEM with a live stack means network reach.
 - *Selective denial.* Asserting resets or powering down domains (PL, GEMs, OCM, APU) is a denial primitive — useful when the defender's response runs on the same SoC.
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10. The Security Posture Summary

`tools/interpret.py` distills §2-§7 into a single **Security Posture Summary** (`rule_security_posture_summary`): one row per control, each with its location, current value, and an OFF/dev → ON/provisioned verdict. Rows shown in **bold** are the ones that most change exposure when ON — RSA enforcement, encrypt-only, JTAG_DIS, AES read-lock, PPK provisioned, secure debug, tamper armed.

Read it first on any new board: all rows OFF means an all-open part; any deviation is the lead to pursue. The exhaustive per-attribute detail behind each row is in [../11-security-attributes.md](#).

11. Future work

- **Hardened-device comparison (the unblocking step).** The payoff is diffing this all-open baseline against a known-hardened ZynqMP — which controls read differently and which capabilities survive each hardening fuse. Acquiring such a part is the highest-value next move.
 - **Live per-core debug-authentication read.** The posture currently *synthesizes* debug authorization from JTAG_DIS/DFT_DIS + SPIDEN/SPNIDEN + DBGEN. A booted-state read of each core’s CoreSight DBGAUTHSTATUS would confirm the live signal (needs core examination, so it stays out of the JTAG-idle baseline to avoid DAP-wedge risk).
 - **AP-topology capture.** Fold `discover.tcl`’s AP enumeration into the report — which APs respond is itself a posture datum (a hardened part may gate the AXI mem-AP while leaving the APB debug AP open).
 - **R5 BootROM dump.** Wake the R5 in booted state and dump from a different bus-master ID than the A53 (the A53-EL3 path is chip-filtered).
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12. References

AMD / Xilinx documentation

- **UG1085** — Zynq UltraScale+ Device TRM (memory map, security model, boot flow, CSU/PMU/eFuse).
- **UG1087** — Register Reference (per-register bit decode).
- **UG1182** — ZCU102 Evaluation Board User Guide (switches, connectors, USB IDs).
- **UG1137** — MPSoC Software Developer Guide (boot chain, FSBL, ATF).

Open-source code (authoritative bit layouts)

- github.com/Xilinx/qemu — the REG32/FIELD register models that are the source of truth for `openocd/lib/zynqmp-regs-qemu.tcl` (`hw/misc/csu_core.c`, `xilinx_zynqmp_crf.c/crl.c/pmu_global.c`, `hw/nvram/xlnx-zynqmp-efuse.c`).
- github.com/Xilinx/u-boot-xlnx — `arch/arm/mach-zynqmp/include/mach/hardware.h` register structs.

- github.com/Xilinx/embeddedsw — FSBL / xilsky / xilsecure address + key-handling definitions.

ARM

- CoreSight Architecture Specification (DAP, APs, DBGEN/NIDEN/SPIDEN/SPNIDEN semantics).
- ARMv8-A Architecture Reference Manual (exception levels, secure-world system registers).

Tooling

- OpenOCD 0.12+ (openocd.org); JEP106 (JEDEC manufacturer IDs for IDCODE decode).

Internal references

- [docs/11-security-attributes.md](#) — the per-attribute catalog behind this volume.
- [docs/05-enumeration-tool.md](#) — the `enumerate.tcl` (18 sections) + `interpret.py` pipeline.
- [openocd/lib/zynqmp-regs-qemu.tcl](#) — verified register/address/bit source of truth.
- [docs/annotations/zynqmp_security.py](#), [docs/findings/zynqmp_rules.py](#) — decoders + posture rules.

End of Volume 3.